

DiscrNet: A Multimodal Hate Speech Detector

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Abstract— Many social media memes convey hate speech by using a complex interplay of visual elements and written content along with sarcasm and culture. [1], [2] Detection of hate speech using this medium is harder than regular text classification due to the fact that such content does not reflect hatred when analyzed individually. This type of content can appear completely normal when considering either the text or the visual alone. DiscrNet, introduced here, is a multi-modal hate speech detector tailored to memes. It incorporates an MMBT-CLIP framework, whereby visual modalities are represented by CLIP ViT-Large-Patch14 and combined with text representations in a binary task of classifying whether a message is hateful or not. [2] In order to detect coded hate, dog whistles, and other potentially harmful discourse which is relevant from the standpoint of policies, the base classifier is extended with Dynamic Policy Gating, which looks up policies semantically similar to a meme in a carefully constructed policy knowledge base. [9], [10], [11] It should be noted that the framework has the possibility to employ a Gemini-based OCR mechanism, which can be used in the case when there is no text provided by a user, and a RAG-based explanation mechanism that can provide policy-driven reasoning for moderation purposes. [6], [9], [10] According to the experiments carried out on the Facebook Hateful Memes development dataset, the MMBT-CLIP base model demonstrates an AUC score of 0.7660, F1-score of 0.7273, and accuracy score of 0.7060. In case of policy-guided gating, the highest-performing model shows an increase in all three metrics: AUC equals 0.7665, F1-score – 0.7314, and accuracy 0.7180.

Keywords — Multimodal Hate Speech Detection; Hateful Memes; MMBT-CLIP; Dynamic Policy Gating; Retrieval-Augmented Generation; OCR; Content Moderation; Meme Classification; Policy-Guided AI.

I. INTRODUCTION

The social media landscape has brought about many changes in regard to how people create and communicate public information. There are very few instances where that communication exists as text alone. Now, social media users frequently mix short statements, screenshots, doctored pictures, icons, and memes to convey an opinion or message briefly and implicitly. While this is helpful when trying to communicate ideas effectively, it makes identifying hate speech and discrimination difficult. Such communications can now be obscured by being included in pictures and text that seem unrelated. [1], [2]

This is made especially hard by the nature of memes. A meme could be made up of a simple picture alongside a caption, but the combination might be derogatory towards someone based on their ethnicity, religion, gender, country, race, or some other protected characteristic. The message may also be conveyed through either the caption or the image. There have been instances where, taken separately, the picture and caption would have little to no meaning at all; but the message becomes hateful once put into context. Previous multimodal hate speech detection studies have indicated that visual cues can play a significant role when detecting hate speech, while the correlation between the two remains tricky for algorithms to understand. [1]

The conventional methods of hate speech identification focused solely on textual data like tweets, comments, and posts. [1], [3] Such systems generally depend on lexicon-based features, word embeddings, or text encoders based on transformers. While these techniques have shown great success in explicit hate speech cases, they are relatively less efficient in identifying

covert intentions of being harmful. [3], [15] Text-only systems face difficulties in meme identification since they overlook any form of visual imagery, facial gestures, contexts present in the image, and the positioning of texts within the image.

The problem of multimodal hate speech detection addresses these issues by analyzing both the image and the text. The recent advances in vision-language models make it possible to create models that learn joint representations for both modalities, which makes them more adequate for classification tasks related to memes. [5], [6], [8], [16] Nevertheless, multimodal classification is far from being solved. [1], [2], [5] The recognition of hate speech in memes requires not only an understanding of images and texts but also specific social backgrounds, encoded meanings of statements, and definitions of harm that are policy-dependent.

In this paper, DiscrNet, which is a multimodal hate speech detection model for memes guided by a specific policy, is proposed. [2], [7], [10] The model utilizes an MMBT-CLIP framework that consists of both text and image representations in terms of binary classification. Image representations are computed using CLIP ViT-Large-Patch14 encoder, whereas textual data is transformed into multimodal representation using the bi-transformer framework for multimodality. [2], [5]

The key new feature introduced by the DiscrNet framework is that of Dynamic Policy Gating. Rather than considering only the classification probabilities produced by the model, the input image is compared to a database of policies regarding safety and harmful narratives. When the strength of the policy signal extracted from the database is sufficient, then it impacts the final prediction via gating. The system is thus able to detect

instances of discrimination, especially those which involve more subtle text, by combining knowledge about safety policies with deep learning models. [7], [9], [10], [11]

Apart from that, DiscrNet contains two components that make it more applicable to real-world applications. First, DiscrNet relies on the OCR module based on the Gemini model. The use of this module becomes critical when there is no caption available to the model. Second, DiscrNet makes use of the RAG-based explanation module that looks up relevant legal background information about the decision made and formulates an explanation in natural language of that decision. [9], [10], [11]

The experiment is done on the Facebook Hateful Memes development dataset and MMHS150K dataset. [1], [2] Without the policy-guiding gating strategy, the MMBT-CLIP model performs with an AUC score of 0.7660, an F1-score of 0.7273, and an accuracy of 0.7060. With the addition of the policy-guiding gating strategy, the highest performing version of the algorithm obtains an AUC score of 0.7665, an F1-score of 0.7314, and an accuracy of 0.7180. While the increase in performance is small, this proves that policy retrieval does provide extra signal to the multimodal classification task.

The main contributions of our work are listed below:

- (1) We built the multimodal hateful meme detector using MMBT-CLIP with CLIP ViT-Large visual embeddings. [2], [5]
- (2) We proposed Dynamic Policy Gating mechanism which controls the classifier output by using policy similarity through retrieval process.
- (3) We used Gemini OCR for detecting memes where image text is not included in the caption provided to us.
- (4) We introduced a new component called RAG Reasoner which is responsible for generating policy grounded explanations for the memes based on retrieval-augmented generative reasoning process. [9], [10], [11]
- (5) We compare experimentally three configurations: base model, policy scorer, and policy gated ensemble models.

The remaining part of this paper is structured as follows. Section II provides literature review on the following topics: text-only hate speech detection, multimodal hate speech detection, CLIP-based meme detection, retrieval-augmented generative reasoning, and ensemble-based moderation systems. Section III gives detailed explanation about dataset used in experiments along with data pre-processing pipeline. Section IV introduces DiscrNet architecture. Section V discusses experiments conducted along with evaluation metrics. Section VI presents experiments findings along with discussion and conclusion on possible limitations.

II. RELATED WORK

Automatic hate speech detection has been investigated for several years, although the problem itself has evolved as online

communication has shifted from textual to more visual content. Early algorithms were focused on textual content, whereas modern algorithms should be able to analyze image-text pairs, especially memes. [1], [2] This section briefly describes the key directions related to DiscrNet.

A. Text-Only Hate Speech Detection

The early models for hate speech detection were generally targeted at very short pieces of text like tweets and forum comments. [1], [3], [15] Most of these models were based on supervised machine learning approaches. In traditional techniques, feature extraction relied heavily on manually created features such as word n-grams, TF-IDF scores, sentiment analysis features, and lexicon-based features. Popular classification algorithms include support vector machines, logistic regression, decision trees, and random forests. [1], [15]

Such approaches worked when the language used in the hate expression included abusive terms and lexical patterns. However, they failed when the context was critical to the understanding of its meaning. Deep learning-based systems made improvements in text categorization through learning rich semantic representation techniques. More advanced architectures based on the transformer model, such as BERT and RoBERTa, have become more widespread as foundations for detecting hate speech due to word representations being based on contextual information. [3], [4], [12], [15]

In spite of these advancements, however, there are limitations with using only text classifiers for meme-based content moderation. For instance, in some memes, the entire meaning is not contained in the text itself. A text-only classifier could either overlook the presence of hate speech or flag content unnecessarily. [1], [2]

B. Multimodal Hate Speech Detection

Multimodal hate speech detection advances this area further by utilizing both visual and textual features. [1], [2], [5], [8], [16] Multimodal approaches have proven crucial for several reasons since posts on social media usually incorporate both texts and visual data in one place. A reference paper by Gomez et al. proposed the MMHS150K dataset and analyzed hate speech detection for multimodal tweets including images and text. [1] One finding from the research was that visual information could add valuable insights, although multimodal models could not always surpass text-based models due to their interplay. [1]

Various approaches have been proposed for the problem of fusing in previous studies. [1], [2], [4], [5], [8], [16] Simple methods simply combine visual features and textual features, while advanced models use attention, bilinear pooling, and transformer-based multimodal encoders. This helps the model to understand the correlation between different aspects of vision and text. [1], [6], [7]

C. Hateful Memes and Vision-Language Models

Facebook's Hateful Memes dataset added complexity to the task of identifying memes. [2] While typical social media datasets consist of instances that can be classified solely on the basis of their image or text content, the Hateful Memes dataset includes instances where understanding both the modalities together is necessary. [2]

Vision-language models are ideal for use in such environments since they capture the association between images and language. [5], [6], [8], [16] CLIP is particularly beneficial because of its training on image-language pairs at scale and its ability to create high-quality visual and semantic features. CLIP ViT-Large-Patch14 is specifically selected as the visual encoder in DiscrNet due to its superior feature extraction capability compared to previous networks like ResNet or Inception. [5], [7]

The MMBT-CLIP approach is used for the system. [2] MMBT is the abbreviation of Multimodal Bi-Transformer. The method lets the fusion of images and textual attributes within a transformer-like model. It would work better than a combination done later since it is easier to consider the interaction between both modalities before making a classification.

D. OCR for Meme Text Extraction

An issue that arises in practice when it comes to classifying memes is the fact that relevant text is often contained within the image. [1], [2], [6] When there is a caption available in the dataset, the caption can simply be used for classification. But in practice, when the user uploads an image but no text is entered, the text needs to be extracted first.

The older multimodal methods often relied on conventional OCR techniques to extract the text from the images. [1] However, the text in memes is generally noisy, stylized, compressed, skewed, or overlayed on complicated backgrounds. In DiscrNet, the OCR technique relies on the Gemini model to enhance accuracy. [6]

E. Retrieval-Augmented Reasoning for Hate Speech Moderation

The use of Retrieval-Augmented Generation (RAG) has increasingly become helpful for problems that require extra information from a model. [9], [10], [11] When it comes to hate speech detection, retrieval may serve several purposes: defining policy regarding whether a certain post attacks a certain protected group, retrieving harmful narratives or code words, and assisting in explaining decisions made by linking a prediction with applicable policies for moderation. [7], [9], [11]

There are two forms of retrieval in the DiscrNet system. First, there is the retrieval of policies in scoring mode, where the query is matched against the selected database of policy

concepts. Second, there is the RAG-driven generation of an explanation, whereby the retrieval result of the policy context is used by Gemini to produce a readable explanation.

F. Ensemble and Policy-Guided Fusion

Ensemble learning uses multiple input signals/models for better robustness. [12], [13], [14], [15] DiscrNet employs an efficient ensemble-based method known as Dynamic Policy Gating. While conventional ensembling techniques entail aggregating results from numerous classifiers, DiscrNet aggregates the output of the base MMBT-CLIP classifier with the corresponding policy signal obtained via retrieval.

G. Research Gap

The literature review reveals that there have been advancements in the areas of hate-speech recognition using text-only data, meme categorization using multimodal data, OCR-based text extraction, and retrieval-based reasoning. [1], [2], [6], [9], [10], [15] Nevertheless, these tasks are usually considered independently. Classifiers usually give predictions without providing policy-related explanations. [9], [10], [11]

DiscrNet solves the problem above by incorporating four key elements into an integrated pipeline: (1) A powerful MMBT-CLIP model for classifying hate memes [2], [5]; (2) Gemini OCR for retrieving meme texts in the absence of captions; (3) Dynamic Policy Gating for fusing classifier probability with policy similarity score; and (4) Explanation generation using a RAG model to generate policy-aligned justification. [9], [10], [11]

III. DATASET AND PREPROCESSING

The system in question was created and tested primarily using the Facebook Hateful Memes Dataset, which is used as a benchmark for binary classification of hateful memes. [2] Each sample within this dataset consists of a combination of an image, text, and the respective label of whether the meme is a hateful meme or not.

A. Dataset Format

The dataset is organized using image files and JSONL annotation files. [2] Each JSONL file contains one meme sample per line. A typical record includes the fields: id, img (path to image), text (meme caption or embedded text), and label (binary 0 or 1). Table I defines the binary label scheme.

TABLE I. Dataset Label Definitions

Label	Meaning
0	Non-hateful meme
1	Hateful meme

For training and evaluation, the system reads the JSONL files, loads the corresponding image, extracts the textual field, and prepares both modalities for the MMBT-CLIP model.

B. Input Modalities

DiscrNet uses three possible sources of information from each meme.

1) Image Content:

The image is the core visual data. It may include humans, symbols, things, screenshots, modified visual content, and meme templates. The image passes through the CLIP-based visual encoder (openai/clip-vit-large-patch14). [1], [2], [5]

2) Caption Text:

Each meme comes with a text field. Captions are crucial because memes tend to carry their negative message in the form of a caption, an insult, joke, stereotype, or reference. [1], [2] Tokenization is performed by the transformer tokenizer for the input text data.

3) OCR-Extracted Text:

During practical usage, the meme can also be posted without a separate caption. [1], [2] For such cases, DiscrNet has an OCR module that allows it to extract text from the image if the caption is not provided by the user. Gemini-based OCR is applied here for the reason that it is better suited for meme texts, since the latter tend to be stylized, compressed, placed against elaborate backgrounds, or arranged in an unconventional way. [6]

C. Image Preprocessing

After processing the image, it is fed to the CLIP encoder after being converted to its format. The preprocessing steps include: loading the meme image from the dataset directory; converting the image to RGB color format; resizing and normalizing the image to satisfy the needs of the CLIP processor; and passing the transformed image to the CLIP ViT-Large encoder. The output of the CLIP encoder will be visual embeddings of the semantics in the meme image. [5]

D. Text Preprocessing

Textual input undergoes preprocessing according to the pipeline suitable for transformers. It includes capturing the caption or the text recognized by OCR; deleting irrelevant or invalid text; tokenizing with the model tokenizer; cutting or padding tokens into the same maximum length; and building attention masks required for processing with transformers.

The model is not expected to perform extensive text cleaning since it can utilize contextual hints encoded in the text. [3], [4] Punctuation marks, capitalization, offensive language, etc., can provide helpful signals for classifying memes. Cleaning may

result in loss of information necessary for the model to capture context correctly.

E. Policy Knowledge Base Preparation

A crucial aspect of DiscrNet is the policy knowledge base. [9], [10], [11] It consists of a curated database containing policies, definitions of harmful content, and discriminatory narratives. Every policy is vectorized and saved to retrieve its similarity with the input meme at inference time. [9], [10], [11]

The knowledge base enables two features: (1) Dynamic Policy Gating, where the retrieved policy similarity value serves as another feature for the output decision of the classifier; and (2) Explanation via RAG, where the policy text is retrieved by the system and fed into the Gemini reasoning engine to generate an explanation for potential hate-speech violation or compliance.

F. Handling Missing Captions

One practical issue in meme classification is missing or incomplete caption text. DiscrNet handles this using the following logic: if caption is provided, use the provided caption as text input; if caption is empty, apply Gemini OCR to extract text from the image; if OCR also fails, use a blank or placeholder text input and rely mainly on the image pathway.

G. Train and Development Split

The system is trained on the training split of the Facebook Hateful Memes dataset and evaluated on the development set. The development set is used to compare: the base MMBT-CLIP classifier; the standalone policy scorer; and the policy-gated ensemble output. The same development split is used across all configurations so that the comparison remains fair.

H. Class Imbalance and Loss Function

Datasets for hate speech detection can be difficult for class distribution and hard examples. [1], [2], [3] While non-hate speech examples might have offensive language not directed towards a particular protected community, hateful memes might use euphemistic terms rather than slurs.

DiscrNet adopts the Focal Loss technique during training to counter such challenges. [2] The Focal Loss technique emphasizes those samples whose labels are incorrect or predicted with high uncertainty. Generally speaking, the loss function helps to reduce attention on examples belonging to the majority class and instead emphasize confusing samples.

I. Final Input Pipeline

The complete preprocessing pipeline begins with the Meme Image and Caption. If the caption is missing, Gemini OCR is applied. The image and text are then separately preprocessed through CLIP ViT-Large Encoder and Tokenizer respectively to produce Visual Features and Text Tokens. These are fused in

the MMBT-CLIP Fusion layer to produce a Base Hate Probability, which then passes through Policy Retrieval and Similarity before Dynamic Policy Gating produces the Final Label along with a RAG Explanation.

IV. PROPOSED METHODOLOGY

DiscrNet is conceived as a multimodal hate speech detection pipeline designed for meme content. The basic assumption of the system is to employ a powerful image-text classifier together with external policy information. The latter becomes a supplementary contextual cue when memes seem to be aligned with hateful stories and violate the existing safety policies. [9], [10], [11]

DiscrNet includes the following four main parts: (1) Inputs preprocessing and OCR; (2) MMBT-CLIP multimodal classification; (3) Dynamic Policy Gating; and (4) RAG-based explanation generation.

A. Overall System Flow

DiscrNet implements a two-tier decision process. At the first tier, a neural-based multimodal classifier is used to determine whether a meme is hate or non-hate. At the second tier, the meme is assessed based on whether it semantically aligns with the existing harmful content guidelines or narratives. The complete system flow proceeds from Input Meme → Preprocessing (CLIP Image Encoding + Text Tokenization) → MMBT-CLIP Fusion Model → Base Hate Probability → Policy Retrieval and Similarity Scoring → Dynamic Policy Gating → Final Prediction → RAG-Based Explanation.

B. Input Layer

The system can receive two possible inputs: (1) an image and its caption, or (2) an image alone. If there is a caption, then it is used directly as the text input by the system. But if there is no caption, then the image itself will be processed through the OCR engine to obtain the visible text from the meme.

C. Gemini-Based OCR Module

DiscrNet employs the OCR module when the caption field does not have any information. Rather than just relying on conventional OCR modules, DiscrNet employs Gemini OCR for text extraction from memes. [6] The logic is: if caption is available, use it directly; if caption is missing, send image to Gemini OCR; if Gemini OCR returns text, use extracted text as the caption; if OCR fails, pass empty text and rely mainly on image features.

D. MMBT-CLIP Multimodal Classifier

The key classifier used in DiscrNet follows MMBT-CLIP architecture. [2], [5] The vision transformer model employed is openai/clip-vit-large-patch14. The CLIP vision encoder converts the meme image into its visual embedding. CLIP vision encoder is appropriate for the task at hand since it has

been trained on image-text pairs, enabling it to capture the visual content in language-aware terms. [5]

Once the text data is extracted either from the dataset caption or the OCR process, it is tokenized and passed along with the image representation into the multimodal fusion layer. This enables the classifier to determine if the combination of text and image is hateful. [1], [2] The classification model generates the base probability P_{base} = probability of the meme being hateful.

E. Efficient Fine-Tuning with LoRA

The training process of VLMs is costly when training is done on limited resources. To allow the model to be trained on consumer-grade GPUs, LoRA adapters were used for efficient fine-tuning. [7] LoRA enables fewer trainable parameters by introducing tiny trainable adapter matrices rather than training all model weights. It makes it possible for the system to fine-tune the pretrained model into hate meme detection tasks without fine-tuning the whole model.

F. Focal Loss for Hard Examples

Detection of hateful memes poses challenges because some memes may have a subtle form of hate speech and lack any overt slurs. There also exist memes that are not considered hateful but have offensive language which does not target any protected class. DiscrNet employs Focal Loss when training the machine learning model. [2] Focal Loss focuses the network's attention on harder instances or instances classified wrongly by the current model.

G. Policy Knowledge Base

The policy knowledge base is a manually curated list of statements on safety policies, hate speech definitions and harmful content narratives. [9], [10], [11] Each policy statement is embedded and saved for matching. In the process of inference, the meme's text will be matched with those policy statements which have been embedded. Policy statements capture ideas such as: attacking the members of a protected group; dehumanization language; hateful stereotypes; discriminatory coding; vulgarity that is not hate speech; and policy differentiating hate from harassment or just being offensive.

H. Policy Retrieval and Similarity Scoring

After the meme text is prepared, the system computes its similarity with the policy knowledge base. [9], [10], [11] The highest similarity score is selected as the policy score. The process is: (1) generate text embedding from input text or OCR text; (2) compare with policy embeddings; (3) retrieve most similar policy; (4) return policy similarity score.

V. EXPERIMENTS

The experiments described in this section aim to assess the effectiveness of DiscrNet's key components: the base multimodal classifier, the policy retrieval module, and the dynamic gating combination.

A. Evaluation Metrics

The experiments are evaluated using four core metrics: Area Under the ROC Curve (AUC), which measures the model's overall ability to distinguish between hateful and non-hateful memes; Average Precision (AP), which summarizes a precision-recall curve and accounts for class imbalance; F1-Score, which balances precision and recall; and Accuracy, which measures overall classification correctness.

B. Evaluation Protocol

For each meme in the development set: (1) produce the initial prediction with MMBT-CLIP; (2) produce the policy similarity score; (3) either use an ensemble or a gating function; (4) save the prediction score; (5) calculate the accuracy with respect to the ground truth label. Once all memes have been evaluated: calculate the AUC, Average Precision, F1-score, Accuracy, and compare all models.

C. Implementation Tools

Table II summarizes the tools and libraries used in the implementation of DiscrNet.

TABLE II. Implementation Tools and Libraries

Purpose	Tool / Library
Deep learning	PyTorch
Transformer models	HuggingFace Transformers
Vision-language encoding	OpenAI CLIP
Efficient fine-tuning	LoRA
OCR	Gemini API
RAG reasoning	Gemini
Vector search / retrieval	FAISS / policy embeddings
Interface	Gradio
Metrics	scikit-learn

D. Experimental Goal

In addition to aiming for the maximum score, the goal of the experiment was to evaluate the utility of policy-aware multimodal moderation. The experiment is concerned with three main questions: (1) What is the performance level of the base MMBT-CLIP classifier? (2) Is there any utility in policy similarity in isolation as a classifier? (3) Does the Dynamic

Policy Gating algorithm augment the performance of the base model?

VI. RESULTS AND DISCUSSION

In this section, the experimental results of DiscrNet on the Facebook Hateful Memes development set are discussed. [2] The aim of the experiment was to compare the baseline multimodal classifier with the policy-based retrieval module and to evaluate whether incorporating both models into one using Dynamic Policy Gating enhances performance compared to the baseline model only.

It was found out that the MMBT-CLIP model demonstrates the best performance among the individual classifiers while the policy scorer alone does not provide a sufficiently accurate score for meme classification. But when used correctly as a component of gating, the proposed technique provides a slight boost in the overall model performance.

A. Quantitative Results

Table III shows the main comparison between the three evaluated configurations.

TABLE III. Quantitative Comparison — Facebook Hateful Memes Dev Set

Model Configuration	AUC	AP	F1	Acc.
Base MMBT-CLIP	0.7660	0.7586	0.7273	0.7060
Policy Scorer Only	0.5192	0.5149	0.6676	0.5080
Policy-Gated Ensemble	0.7665	0.7611	0.7314	0.7180

The baseline MMBT-CLIP model obtains an AUC score of 0.7660, which demonstrates its ability to distinguish hateful and non-hateful memes adequately. The high F1-score (0.7273) and accuracy (0.7060) prove the quality of the multimodal fusion model. The performance of the policy scorer as the stand-alone classifier is much worse, with AUC score equal to 0.5192 and accuracy of 0.5080, meaning that classification based on policy similarities is nearly random. The most effective solution is the policy-gated ensemble approach, improving F1-score from 0.7273 $\rightarrow$ 0.7314, accuracy from 0.7060 $\rightarrow$ 0.7180, AUC from 0.7660 $\rightarrow$ 0.7665, and AP from 0.7586 $\rightarrow$ 0.7611.

B. Base MMBT-CLIP Performance

The base model of MMBT-CLIP works efficiently since it employs both textual and visual data. [2], [5] Using CLIP ViT-Large-Patch14 is advantageous due to the ability of this model to generate high-quality visual features. [5] Nevertheless, the base model can be challenged by hateful memes that rely on hidden stereotypes, coded language, allusions to politics and culture, sarcasm, group targeting in an indirect manner, or image-text contradiction.

C. Standalone Policy Scorer Performance

The policy scorer performs poorly on its own. An AUC of 0.5192 suggests that the similarity between policies and memes alone is insufficient to distinguish between hateful and non-hateful images. There are multiple reasons for this finding. First, the policy scorer heavily relies on textual similarity. [9], [10], [11] Second, legal policies employ more formal language than memes. Third, a similarity-based search might yield irrelevant results.

D. Effect of Dynamic Policy Gating

In Dynamic Policy Gating, the base model is enhanced in the way that the policy signal impacts the output of the model only when the similarity score is high. The logic is: when the policy match is poor, rely on the MMBT-CLIP classifier; when the policy match is good, boost the probability of hateful content. By improving the F1-score from 0.7273 to 0.7314, it is seen that the gated model better handles the balance of precision and recall. [10], [12], [15]

E. Dynamic Gating Variant

A second dynamic analysis was also performed using an optimized gating setup. Table IV presents the results.

TABLE IV. Dynamic Policy Gating Variant — Ablation Results

Model Configuration	AUC	AP	F1	Acc.
Base MMBT-CLIP	0.7660	0.7586	0.7273	0.7060
Policy Scorer Only	0.5192	0.5149	0.6676	0.5080
DPG Variant	0.7663	0.7596	0.7276	0.7140

This modification increases accuracy from 0.7060 to 0.7140; however, the increase in the F1-score is very minimal. The current modification shows weaker performance compared to the previous model using policy gating in F1-score and AP. Therefore, the most optimal parameters recorded are: AUC = 0.7665, AP = 0.7611, F1-score = 0.7314, Accuracy = 0.7180.

F. Practical Value of RAG-Based Explanation

The explanation provided by the RAG-based layer is not evaluated based on metrics such as AUC or F1-score since it does not play the role of the primary classifier. [9], [10] With the help of the RAG module, the DiscriNet gains additional usefulness as an aid for moderators since it provides them with the context of the policies and a brief explanation as well. [9], [10], [11]

G. Error Patterns and Limitations

1) Implied Hate:

The memes containing hate in an implied way should be identified. Such memes make use of humor, stereotyping, and cultural knowledge; hence, the DiscriNet model might fail to

identify all of these examples since the caption as well as the policy entry do not convey the hateful sentiment. [1], [6], [7]

2) Dependence on Text:

The policy scoring procedure used in the process is quite text-based. [9], [10], [11] As a result, if the harm conveyed is based primarily on visuals, policy search will be of no help.

3) Policy Search Noise:

Sometimes, the policy entry obtained from the search may have some connection to the topic mentioned in the meme without conveying any hateful message in the context. High scores for such memes might cause false positives.

4) OCR Misreadings:

Gemini OCR enhances text recognition, but OCR is not flawless. In case of misreading by OCR, the classifier as well as the policy retrieval unit will receive incorrect data.

5) Data Collection Scope:

In the present experiment, data collection was concentrated around the development datasets, Facebook's Hateful Memes and the MMHS150K. Thus, the system does not guarantee universal applicability to different social networks, linguistic systems, or meme types. [1], [2], [11]

H. Results Summary

The results of the experiment yield three main findings. Firstly, MMBT-CLIP acts as a very strong base model for hateful memes classification. [2], [5] Secondly, although not too effective on its own, the feature of policy similarity is useful as a complementary signal to the core classifier. Thirdly, Dynamic Policy Gating leads to the best final performance among all considered approaches. [10], [12], [15]

VII. CONCLUSION

DiscriNet has been proposed in this work as a policy-driven, multi-modal hate speech detection system that can detect hate speech in memes. [1], [2], [10] DiscriNet focuses on an important problem faced by most hate speech classifiers where memes that are considered offensive may not make sense when only looked at in isolation.

The primary architecture of DiscriNet uses MMBT-CLIP as its backbone and consists of CLIP ViT-Large-Patch14 to represent visual input and the text component which comes from the caption present in the dataset, or if not available, from Gemini-based OCR. [2], [5], [6] The Dynamic Policy Gating layer introduces greater sensitivity towards policies into the system. [9], [10], [11], [15]

The experimental results demonstrate that the baseline MMBT-CLIP classifier shows good results for the development set of the Facebook Hateful Memes dataset — 0.7660 AUC, 0.7273

F1-score and 0.7060 accuracy. On the other hand, the standalone policy scorer produces poor results — 0.5192 AUC. Using the Dynamic Policy Gating technique, the best configuration improves the scores up to 0.7665 AUC, 0.7314 F1-score and 0.7180 accuracy.

These results demonstrate that policy-driven retrieval can bring a slight yet valuable boost compared to the stand-alone multimodal classifier. Additionally, the system becomes more interpretable as predictions are linked with the reasoning process using policies. [9], [10], [11] Thus, the DiscrNet architecture shows that detection of hateful memes can be performed better through integration of: (1) image-text multimodal classification; (2) retrieval and gating based on policy; and (3) generation of explanations for human review. [9], [10], [11]

A. Future Directions

1) Cross-Dataset Evaluation:

The model should be evaluated using other datasets like HateXplain, HASOC, OLID, and novel multimodal hate speech datasets. [1], [3], [11], [16] Table V outlines the proposed evaluation framework.

TABLE V. Proposed Cross-Dataset Evaluation Framework

Evaluation Type	Purpose
In-dataset testing	Measure performance on the same benchmark
Cross-dataset testing	Measure generalization to new data
Platform-based testing	Compare meme styles across platforms
Language-based testing	Check multilingual robustness

2) Better Policy Knowledge Base:

There are opportunities to enrich and organize the current policy knowledge base. Possible additions might include specific platform hate speech policies, definitions of legality and moderation, examples of dog whistles and coded hate speech, definitions of protected classes, templates of harmful stereotypes, and positive examples of offensive but not hateful speech.

3) Visual Policy Retrieval:

The current policy retrieval process is largely text-based. [9], [10], [11] The future version could incorporate the ability to extract the policy context based on text and image features, making the policy module more relevant when the signal comes from visuals instead of text. [5], [7], [16]

4) Improved Fusion Approaches:

Some possibilities for future development include: meta-learning classifiers; calibration-based fusion; weighting based on confidence; different gates for text, images, and policy input; and estimating uncertainty prior to policy boost.

5) Human-In-The-Loop Moderation:

In view of the sensitivity of detecting hate speech, future iterations need to incorporate feedback from human users. [3], [9], [10] The moderators can choose to accept, reject, or even edit explanations generated by the system, which will be useful for updating the policy knowledgebase and retraining the gatekeeping mechanism.

B. Concluding Remarks

Based on DiscrNet, policy-guided multimodal hate meme detection appears to be a feasible direction. [1], [2], [7], [10] The presented findings confirm that when a good multimodal classifier is available, it may be helpful to integrate it with a well-controlled policy retrieval signal. [9], [10], [12], [15] Hence, the key strengths of the system lie in: slightly better classification accuracy compared to the initial MMBT-CLIP model; and a more interpretable mechanism due to combining the predictions with the retrieved policies.

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